COMPAS 2026

IEEE 3rd International Conference on Computing, Applications and Systems

SMTPS: securing email transmission with SMTP over TLS

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Paper ID: [000]

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Presentation Outline

01 Introduction and research problem

02 Related work and research gap

03 Proposed solution: SMTPS

04 Protocol comparison and evaluation

05 Conclusion

Introduction and Research Problem

Plain SMTP sends mail and passwords in cleartext, where attackers can read or alter them.

01 Email carries credentials and private data across networks

02 Plain SMTP and optional STARTTLS can leave sessions exposed

03 Objective: explain and compare SMTPS with STARTTLS and plain SMTP

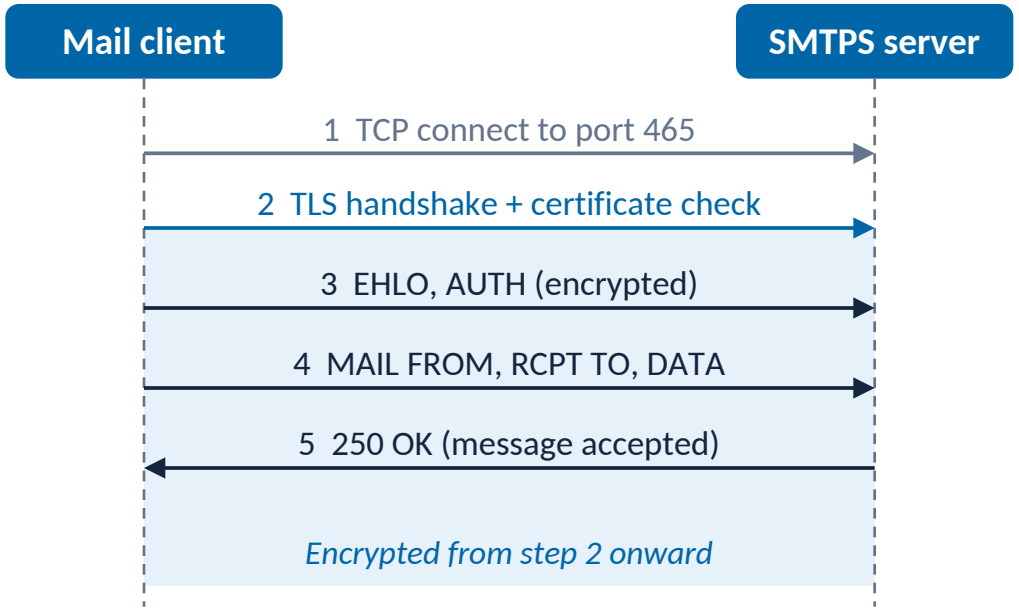
Related Work and Research Gap

Study / approach	Key idea	Relevant limitation
RFC 5321 [1]	Defines SMTP mail transfer	No built-in encryption
RFC 3207 [2]	Upgrades a session with STARTTLS	Cleartext phase can be stripped
RFC 8314 [3]	Prefers implicit TLS (port 465)	Guidance, not a measured comparison

Research gap No concise comparison of implicit TLS (SMTPS) and STARTTLS for mail submission

Methodology

SMTPS: SMTP over implicit TLS (port 465)



01 **[Step 1: preparation**

Server holds a CA-signed X.509 certificate; client trusts the CA

02 **[Step 2: core procedure**

TLS handshake first, then EHLO, AUTH and mail commands inside the tunnel

03 **[Step 3: output**

Server replies 250 OK and relays the message; client checks certificate name

Evaluation Setup

Protocol variants	Plain SMTP (25), SMTP + STARTTLS (587), SMTPS (465)
Fair comparison	Same server, certificate and mail client in each mode
Evaluation criteria	When encryption starts, round trips, stripping exposure
Test procedure	Inspect sessions with openssl s_client and a packet capture

Handshake Comparison

STARTTLS (port 587)	SMTPS (port 465)
1 TCP connect	1 TCP connect
2 220 greeting, cleartext	2 TLS handshake
3 EHLO, cleartext	3 220 greeting (encrypted)
4 STARTTLS command, cleartext	4 EHLO (encrypted)
5 TLS handshake	5 AUTH, mail (encrypted)
6 EHLO again (encrypted)	
7 AUTH, mail (encrypted)	

Fig. 1. Session set-up order. Orange = cleartext, blue = encrypted. Shown for client submission.

Main finding

STARTTLS starts in cleartext, so an attacker can try to strip the upgrade.

SMTPS has no such phase: encryption begins before any SMTP command.

Protocol Comparison

Method	Typical port	Encryption starts	Stripping risk
Plain SMTP	25	Never	Not applicable
SMTP + STARTTLS	587	After STARTTLS	Yes, if not enforced
SMTPS (implicit TLS)	**465**	**At connection**	**No cleartext phase**

SMTPS encrypts from the first byte, so a network attacker has no cleartext phase to strip.

Conclusions and Future Work

SMTPS is the simplest way to guarantee encrypted mail submission.

01 Contribution: a clear comparison of plain SMTP, STARTTLS and SMTPS

02 Limit: protects the hop only, not end-to-end (S/MIME, PGP)

03 Next: measure latency and deployment; add MTA-STS for server links

Selected References

[1] J. Klensin, “Simple Mail Transfer Protocol,” RFC 5321, IETF, 2008.

[2] P. Hoffman, “SMTP Service Extension for Secure SMTP over TLS,” RFC 3207, IETF, 2002.

[3] K. Moore and C. Newman, “Cleartext Considered Obsolete: Use of TLS for Email Submission and Access,” RFC 8314, IETF, 2018.

[4] D. Margolis et al., “SMTP MTA Strict Transport Security (MTA-STS),” RFC 8461, IETF, 2018.

Thank you

Questions and Discussion

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